

Introduction

This report presents recommendations from a proof-of-concept study on whether predictive modelling can identify, at registration, which new clients will make a first deposit. Because STADIOEquities' own client data is sensitive, the approach was first tested on a comparable public

dataset, the UCI Bank Marketing data, where the outcome is whether a client subscribes to a term deposit. The report covers model selection, possible improvements, the adaptations needed for STADIOEquities' own data, and how the findings compare with the published literature.

1. Which model is best, Model 1 or Model 2?

Two models were built and tested on the same 7,880-client test set, which is a later slice of the data so that it mimics scoring genuinely new clients. Model 1 is a Logistic Regression and Model 2 is a Random Forest. Model 2 performed better on every measure: ROC-AUC of 0.71 against 0.53, recall of 0.81 against 0.47, and F1 of 0.55 against 0.39. McNemar's test, which compares the models on the clients where they disagree, gave a p-value below 0.001 (Model 2 was right and Model 1 wrong for 1,575 clients, against 1,246 the other way round), and the 95% confidence intervals for ROC-AUC do not overlap (0.70 to 0.73 against 0.51 to 0.54). We therefore recommend Model 2. Its weakness is precision: only 41% of the clients it flags go on to deposit, and it correctly identifies fewer than half of the clients who do not. It should therefore be used to prioritise outreach, not to exclude clients. Model 1 remains useful as an explainable benchmark. In practical terms, Model 2 finds 2,023 of the 2,514 clients who went on to deposit, against 1,189 for Model 1, but it also flags 2,865 clients who did not deposit.

2. How can the selected model be improved?

Five steps would improve Model 2. First, the decision threshold should be set using the cost of contacting a client and the value of a first deposit, rather than the default cut-off, which currently favours recall over precision. If outreach is cheap, a low threshold that catches most depositors is sensible; if it is costly, a higher one reduces wasted contacts. Second, its settings were fixed in advance, so a systematic search using time-ordered cross-validation should be run. Third, the proxy dataset held only demographic, contact and economic fields, whereas STADIOEquities also has onboarding and app or web behaviour, such as session counts and form completion. These signals measure engagement directly and are likely to be more predictive. Fourth, feature importance or SHAP values would explain individual scores, narrowing the interpretability gap to Model 1. Finally, the proxy data drifted strongly over time, with 6.6% deposits in the training period and 31.9% in the test period, so any live model needs monitoring and a regular retraining schedule. Class imbalance was handled by class weighting; oversampling such as SMOTE (Abidin et al., 2025) could also be compared, as could other tree ensembles such as XGBoost, the best model in Setiyani et al. (2022).

3. How must the selected model be adapted to the requested SS1 data?

Several changes are needed for the SS1 data. Demographic and acquisition fields map fairly directly, but onboarding and behavioural fields need new features, following the same rules: flags for placeholder values and encoding fitted on training data only. Any field filled in only after a deposit, such as a deposit timestamp, must be removed, as call duration was here. The evaluation should again use a time-based split, because acquisition channels and client mix will shift. STADIOEquities' class balance (about 59% deposit and 41% none) is far milder than the proxy data's, so class weighting may be unnecessary and the threshold must be set again. Because the proxy data describes bank telemarketing, its patterns will not transfer; the pipeline should be reused, but the model must be retrained on STADIOEquities' data. Finally, the work must move to an access-controlled environment and be reviewed for compliance with data-protection and

financial-services rules before any live use. Results should again be reported with confidence intervals and a significance test, so that STADIOEquities can see whether a difference between

models is real.

4. Do the results align with the related literature?

The results broadly agree with the literature. Setiyani et al. (2022) found that tree-based ensembles beat logistic regression on this dataset; XGBoost performed best there, and our Random Forest is another tree ensemble showing the same direction. Moro, Cortez and Rita (2014) showed that call duration leaks the outcome and evaluated models on time-ordered data. Our modest absolute scores, with accuracy of 57% against the 90% or more reported by Setiyani et al. (2022) and Abidin et al. (2025), likely reflect the stricter design used here: no duration field and a later test period. Abidin et al. (2025) also found that handling imbalance and tuning improved results. We used class weighting only and no tuning, so both remain open gains. They also stressed minority-class recall, which is why we report recall, precision and ROC-AUC alongside accuracy. Overall, the pattern matches the literature, which supports transferring the approach to STADIOEquities' data.

Conclusion

We recommend that STADIOEquities proceeds to the next stage using Random Forest as the primary candidate, with Logistic Regression as a benchmark, subject to the data access, governance and leakage checks above. The proxy study shows that the pipeline works from start to finish and that a non-linear model adds real value. The true level of predictive power can only be established on STADIOEquities' own data.

References

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